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Studierichting: Informatica
Oefeningenreeks 4A nr. 57.1

$$I = \int \frac{\sin x}{2\cos x + \sin x} dx$$

$$\int \frac{\frac{\sin x}{\cos x}}{\frac{2\cos x + \sin x}{\cos x}} dx = \int \frac{\tan x}{2 + \tan x} dx$$

$$\overset{\tan x = t}{=} \int \frac{t}{2+t} \frac{dt}{1+t^2} = \int \frac{(2+t) - 2}{2+t} \frac{dt}{1+t^2}$$

$$= \int \frac{1}{1+t^2} dt - 2 \int \frac{1}{(2+t)(1+t^2)} dt$$

$$= \operatorname{Bgtan} t - 2 \left( \int \frac{A}{2+t} dt + \int \frac{Bt + C}{1+t^2} dt \right)$$

$$\Rightarrow \frac{A(1+t^2) + (Bt+C)(2+t)}{(2+t)(1+t^2)} \Rightarrow (A+B)t^2 + (2B+C)t + (A+2C) = 1$$

$$\Rightarrow \begin{cases} A + B = 0 \\ 2B + C = 0 \\ A + 2C = 1 \end{cases} = \begin{cases} B = -A \\ -2A + \frac{1-A}{2} = 0 \\ C = \frac{1-A}{2} \end{cases} = \begin{cases} A = \frac{1}{5} \\ B = -\frac{1}{5} \\ C = \frac{2}{5} \end{cases}$$

$$\Rightarrow I = \operatorname{Bgtan} t - \frac{2}{5} \int \frac{1}{2+t} dt - 2 \int \frac{-\frac{t}{5} + \frac{2}{5}}{1+t^2} dt$$

$$= \operatorname{Bgtan} t - \frac{2}{5} \int \frac{1}{2+t} d(2+t) + \frac{2}{5} \int \frac{t}{1+t^2} dt - \frac{4}{5} \int \frac{1}{1+t^2} dt$$

$$= \operatorname{Bgtan} t - \frac{2}{5} \ln|2+t| + \frac{\cancel{2}}{5} \int \frac{\cancel{t}}{(1+t^2)\cancel{2t}} d(1+t^2) - \frac{4}{5} \operatorname{Bgtan} t$$

$$= \frac{1}{5} \operatorname{Bgtan} t - \frac{2}{5} \ln|2+t| + \frac{1}{5} \ln|1+t^2| + C$$

$$= \frac{1}{5} \left( x - 2\ln|2 + \tan x| + \ln|1 + \tan^2 x| \right) + C$$

# References